BP-based intelligent case retrieval method of high-rise structure form selection

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Abstract

Establishing a case retrieval algorithm is the key step of the high-rise structure intelligent form optimization (IFO) based on Case Based Reasoning (CBR). First, analyzes the defects of the traditional case retrieval method (CRM) based on typical distance and the characteristics of architecture structure case retrieval, an improved BP-based intelligent CRM was established and the topological structure and algorithm of the neural network was presented. Next, based on four target cases and twenty-five source cases, the training and forecast of the improved BP-based case retrieval was provided. Finally, the engineering application was given. The result shows: this intelligent CRM can overcome such shortcomings as poor learning ability, low searching efficiency and intelligent level, etc., providing comparatively more scientific method for the CRM of high-rise structure IFO.

1. Introduction

High-rise structural form selection optimization is an important strategic problem concerning synthetic performance of building structure. It possesses some characteristics such as multi-goal, multi- method, multi-solution and high intelligence, etc. So Intelligent design is the direction in the development of form optimization design, knowledge-based intelligent designing system is the chief means of it and knowledge acquirement is the "bottleneck". However, the past two decades saw no breakthrough in either research or application in this area. One big reason is that the solution to the knowledge acquirement of intelligence design remains far from satisfactory. People have accumulated a great number of high-rise engineering cases, which are available or easily acquirable but not costly. CBR is one of the important approaches to solving the "bottleneck" problem of acquiring the knowledge about high-rise structure IFO

CBR is a new important reasoning technique of artificial intelligence. CBR search the case library for case sets similar to the current issue. After evaluation, part or all of them will be reused after amendment. The process of the CBR includes: acquirement and expression of cases, ascertainment of target cases, cases search, cases evaluation, selection optimization, reuse, modification and learning[1]. During the process, case retrieval algorithm design is the key of CBR.

The usual case retrieval methods are the distance-based method, the nearest neighbor one and the induction one[2,3] etc.. The above mathematical analysis-based methods can express directly and clearly similarities between cases. But they cannot do self-learning and their intelligent lever is low. Moreover, to the cases that are difficult to be expressed in mathematical way, we cannot expect a desirable result with those methods. However, with their past knowledge and designing experience and through recollection and association, even the experts don not strictly follow the mathematical expression, still they are able to identify and find some similar cases from the data in hand. Therefore, to enhance the intelligent level of case retrieval, this paper will establish a BP-based intelligent CSM.

2. The Improved BP-base Intelligent CRM

2.1 Case Similarity Measure & Mixed CRM.

2.1.1 Ascertainment of Target Cases,First, CBR needs to distinguish the characteristic attributes of the significant cases to form selection and then transforms them into target cases. After that, it does search work according to them. The initially ascertained target case

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attributes vector is $Y=\{y_k\}_m=\{$constructing performance, construction height, floor, fortification intensity, site class, basic wind pressure, height-to-width ratio and length-to-width ratio $\}_8$. The case library to be used in this paper is $X=\{X_i\}_n=\{x_{ik}\}_{n\times m}$ (For the establishment of it see paper [4]), where x_{ik} is the kth attribute value of X_i.

2.1.2 Classification of Case Attributes and the Similarity Measure Method, (1) Attribute classification of high-rise structure cases: the attributes of high-rise cases are divided into four categories in general: enumerating quality attributes (e.g. constructing performance), enumerating quantitative attributes (e.g. fortification intensity and site class, etc.), continuous quantitative attributes (e.g. constructing height and floor, etc.) and fuzzy attributes (e.g. plane regularity and construction convenience, etc.).

(2) Similarity measurement of the attributes of high-rise structural cases: d_{ik}, the measurement value of similarity between the kth attribute of case X_i, x_{ik} and the relative attribute of the target case y_k can be basically ascertained as follows: when d_{ik} is zero they are the most similar and when d_{ik} is one they are the least similar.

Similarity measurement between enumerating quality attribute

$$d_{ik}=\begin{cases}0 & \text{when the attribute values of } x_{ik} \text{ and } y_k \text{ are the same}\\ 1 & \text{when the attribute values of } x_{ik} \text{ and } y_k \text{ are different}\end{cases} \quad (1)$$

Similarity measurement between the enumerating and continuous quantitative attributes:

$$d_{ik}=|x_{jk}-v_k| \quad (2)$$

Similarity measurement between the fuzzy attributes: fuzzy attributes can be expressed with fuzzy numbers. expression (3) gives the triangle fuzzy number corresponding to k^{th} attribute x_{ik}, where x 、x_{ik} and d_i are respectively the kth attribute variable, the triangle fuzzy number barycenter and the half width of interval. Now the fuzzy number of y_k is $\mu_{yk}(x)$, the similarity between the fuzzy number can be ascertained by expression (4).

$$d_{ik}=\ _x\ _R(\mu_{ik}(x)\quad \mu_{yk}(x))=|x_{ik}-y_k|/(2d_i) \quad (3)$$

$$\mu_{ik}(x)=\begin{cases}1-|x-x_{ik}|/d_i & x\in[x_{ik}-d_i,x_{ik}+d_i]\\ 0 & the\ \ others\end{cases} \quad (4)$$

2.1.3 The Mixed CRM Based on Relative Euclidean Weight Distance, The similarity measure method based on typical distance is incapable of describing weight. Therefore, when dimensions between attributes are not in uniform and their distances are widely different; the part of the attributes in depicting the general similarity of the cases cannot be given full play. To solve this problem and to be able to take into consideration everything important and necessary such as concurrence of quality, quantitative and fuzzy attributes and difference in dimension, this paper established a new similarity measure method based on the relative Euclidean weight distance of attributes (see expression (5)). The result shows that this method can solve the problem of the case retrieval with mixed attribute characteristics

$$Sim_i(X_i,Y)=1-\sqrt{\sum_{K=1}^{m}(w_k d_{ik})^2} \quad (5).$$

$$d_{ik}=|x_{ik}-y_k|/max(|x_{k\,max}-y_k|,|x_{k\,min}-y_k|) \quad (6)$$

In the above expression, $W=\{w_k\}_8$, the vector of the attribute's weight, can be ascertained by consulting experts. d_{ik} is the relative distance between y_k, where y_k and x_{ik} is the kth attribute of Y and X_i respectively. To an enumerating quality attribute, d_{ik} is ascertained by expression (1); to a continuous or enumerating quantitative attribute, d_{ik} is ascertained by expression (2) and to a fuzzy one, by expression (3). x_{kmax} and x_{kmin} are respectively the maximum and minimum values of $\{x_{ik}\}_{Tn}$, the column vector of the kth attribute of all the source cases.

On the basis of the above established method, if we scan the case library x, we will get case sets that meet the similarity threshold value. Next we can ascertain the relative scheme set of form candidates for form selection.

2.2 The Improved BP-base Intelligent CRM

2.2.1 Ascertainment of the Topological Structure of BP Network for Case Retrieval, There are only some experimental conclusions concerning ascertainment of topological structure of networks instead of mature and systematic theory. Next we shall establish the topological structure of the BP network according to the past information and the need of case retrieval.

(1) Ascertainment of the input layer: the function of the input layer is inputting vectors $\{x_{ik}\}_m$ and $\{y_k\}_m$ (k [1,m] 、m m_c 、m_c is the upper limit of m.) of the two case samples Xi and Y. To be able to make dynamic similarity comparisons of cases under various circumstances, we adopt the following strategy: enable m to make stochastic dynamic changes within its limited range , design the input layer node as $q=2m_c$, Node q+1 as a biased one , whose value is always 1. When $m<m_c$, the $2(m_c-m)$ nodes that do not have input parameters are all given the input 0.

(2) Ascertainment of both the hidden layer and the number of nodes: Choice of node number of the hidden layer is usually concerned with the numbers of input

nodes, output nodes and learning samples and learning error, etc. If there are few nodes, the network is lacking in precision and fault-tolerance overlooking errors; if there are too many nodes, the network's learning process will be very long and its generalization poor, so there must be a most suitable node number. In this paper we chose $q=2m_c=20$ hidden layer nodes after trial and error, including the biased node q+1.

(3) Ascertainment of output layer: The output layer is two single nodes obtained by inputting case similarity measure values.

Thereupon we have built the multi-input-and-single-output, three-layer BP network.

2.2.2 Activation Function of Nerve Cells of the BP Network, The activation functions in various shapes greatly influence astringency of the whole network. Tianping Chen[5] and Hornik have proved that any bounded non-constant continuous function can be used as an activation one. Experts like Mhaskar have proven that as long as we give a limit to the amplitude at the infinitude point of a continuous function, any non- polynomial can be regarded as an activation function. We adopted in this paper the most widely used S-shape activation function. Expression (7) gives the activation function of the ith nerve cells in the lth activation layer.

$$f(netl_i) = 1/(1+e^{-netli}) \tag{7}$$

Where $netl_i=wl_i*al-bl_i$ is the input of the ith nerve cells in the lth activation layer; $wl_i=\{wl_{ij}\}_q$ is the joining weight vector between it and the nerve cell q in the preceding layer; wl_{ij} is the joining weight between it and the nerve cell j in the preceding layer; $al=\{al_i\}_{q\times 1}$ is the input vector of the lth layer and $bl=\{bl_i\}_{q\times 1}$ the threshold value vector of it.

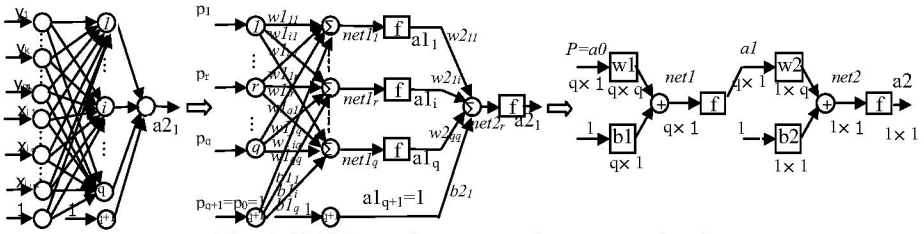


Fig. 1. BP Network structure for case retrieval

2.2.3 Learning Algorithm and Process of the Improved BP Network, The improved BP algorithm is a tutored learning one, the training sample set is $\{X_t,O_t\}_s$, where $X_t=\{x_{tr}\}_q$,$O_t=\{o_{tk}\}_n$,(in this paper) q=20,n=1. Its learning algorithm and process are as follows:

(1) Initialize weight value w1,w2 and threshold values b1, b2, which are the random numbers of the interval[-1,1]

(2) Input a sample $X_t=\{x_{tr}\}_q$ (t [1,s]), increase a biased node and the input vector of it is noted as $IP_t=\{x_{t1},\cdots, x_{tr},\cdots, x_{tq},1\}=\{x_1,\cdots, x_r,\cdots, x_q,1\}$. The expected output is $O_t=\{o_{tk}\}_n$.

(3) Calculate in turn the output vector al of the activation layer l (in this paper l=1,2) and transmit IPt the in forward direction throughout the network, where the output of l=1 equals the input of layer l=2. So the outputs of the nodes of each layer respectively:

$$al_i = f(\sum_{r=1}^{q} wl_{ir}x_r - bl_i) = f(netl_i) \tag{8}$$

$$a2_k = f(\sum_{i=1}^{q} w2_{ki}al_i - b2_k) = f(net2_k) \tag{9}$$

(4) Error control: For the average errors of all samples please see expression (10) and the error of every single sample see expression (11). If it meets the condition E [E] (system's average error limit), et [e] (single sample average error limit) or reaching the set iteration step, learning comes to an end. Otherwise, reversed error transmission will be carried out to turn to step 6.

$$E = \frac{1}{s}\sum_{i=1}^{s} e_i \tag{10}$$

$$e_t = \frac{1}{2}\sum_{k=1}^{n}(o_{ik} - a2_k)^2 \tag{11}$$

(5) Using the output vector a2 and the target vector O_t, calculate $\delta 2_K$ the correction factor of all weight values of the output layer according to expression (12).

$$\delta 2_k = (o_{tk} - a2_k)a2_k(1-a2_k) \tag{12}$$

(6) Correct the v time iteration weight value $w2(v+1)_{ki}$ and threshold value $b1(v+1)$ of the output layer (l=2) according to expression (13) and expression (14). μ and η are respectively the inertia impulse and the learning step factor introduced for the improvement of the traditional BP network.. η decides the changing speed of weight value because if it changes too fast it will lead to local minimum. η takes a value between 0.3 and 0.6 or it may otherwise be ascertained by adopting the

variable step expression in (15), where $\triangle e=e(v)-e(v\text{-}1)$. M and L are respectively numbers greater than or equal to 1 (for instance, M=1.05, L=0.95). μ bigger than η can effectively keep in check or keep within limits the network vibration, preventing the possibility of local minimum. In general, we choose a value between 0.3 and 0.9. The threshold value is corrected according to the weight value.

$$w2(v+1)_{ki} = \mu w2(v)_{ki} + \eta \delta 2_k a1_i \quad (13)$$

$$b2_k(v+1) = b2_k(v) + \eta \delta 2_k \quad (14)$$

$$\eta = \begin{cases} M\eta & \mu = \mu, \quad \Delta E < 0 \\ L\eta & \mu = 0, \quad \Delta E > 0 \end{cases} \quad (15)$$

7) Calculate the correct factor $\delta 1_i$ of all weight values of the hidden layer (l=1) using expression (16).

$$\delta 1_i = (\sum_{k=1}^{n} \delta 2_k w2_{ki}) a1_i (1 - a1_i) \quad (16)$$

8) Correct weight value $w1(v+1)_{ir}$ and threshold value b1(v+1) of the hidden layer(l=1) in accordance with expression (17) and expression (18).

$$w1(v+1)_{ir} = \mu w1(v)_{ir} + \eta \delta 1_i x_r \quad (17)$$

$$b1_i(v+1) = b1_i(v) + \eta \delta 1_i \quad (18)$$

9) Turn to step2. Repeat the above steps with a new sample set $\{X_{t+1}, O_{t+1}\}$ until it satisfies the requirement.

Train the network until it gets ready and then we can use it to do the case retrieval of high-rise structure intelligent form selection.

3.The Training of BP-Based Case Retrieval and its Engineering Example

3. 1 The Training of BP-Based Case Retrieval

Taking the twenty-five cases given in paper [6] as source cases, A1=(49.7, 12, 2, 3, 2), A2=(59.8,20,3,4,2), A3=(89.6,22,1,2,3), A4=(72.1,20,2,2,2) as the target cases to be trained and A5=(61,18,1,3,2) the target case to be searched, we list in Table 1 the similarity measure between the source cases and the target cases ascertained with the expression (5), and using them as learning or forecasting samples to do the NN training and forecast. The forecast result is listed in column forecast value. The average relative error between the foretold value by the neural network and the actual value is only 0.002. It's clear that this method can provide intelligent and highly effective CRM for high-rise structure form selection optimization. Figure 2 shows the NN training process and result.

Table 1 BP network training samples and forecast result

Source cases	Training goal cases			Searching goal cases		Engineering cases	Source cases	Training goal cases			Searching goal cases		Engineering cases
code	*A1*	*A2*	*A3*	*A4*	Forecast	*A5*	code	*A1*	*A2*	*A3*	*A4*	Forecast	*A5*
1	0.7761	0.8160	0.6798	0.6261	0.6271	0.6869	14	0.6494	0.6802	0.7352	0.7749	0.7762	0.7326
2	0.9837	0.8155	0.7524	0.6181	0.6188	0.7816	15	0.6288	0.6564	0.7861	0.6785	0.6794	0.6434
3	0.7990	0.8361	0.6940	0.6242	0.6230	0.7012	16	0.6457	0.6517	0.6432	0.7404	0.7392	0.7204
4	0.7153	0.6761	0.9022	0.5728	0.5733	0.7040	17	0.6607	0.7608	0.6772	1.0000	0.9846	0.7457
5	0.7134	0.7002	0.7961	0.6806	0.6797	0.7964	18	0.6394	0.6258	0.5757	0.5793	0.5800	0.5877
6	0.7090	0.7097	0.9937	0.6512	0.6520	0.7162	19	0.7171	0.6819	0.7759	0.6431	0.6423	0.7843
7	0.6978	0.7146	0.9528	0.6832	0.6821	0.7113	20	0.7989	0.7713	0.7846	0.6240	0.6248	0.7118
8	0.7935	0.7631	0.8965	0.6856	0.6845	0.7990	21	0.7978	0.7561	0.7688	0.7413	0.7403	0.9616
9	0.7935	0.7634	0.7749	0.7585	0.7572	0.9812	22	0.7155	0.6971	0.9437	0.6175	0.6185	0.7137
10	0.7433	0.7824	0.7683	0.7095	0.7105	0.6920	23	0.9131	0.8905	0.7661	0.7378	0.7364	0.7933
11	0.6796	0.7071	0.8963	0.7023	0.7033	0.6964	24	0.7069	0.7955	0.7169	0.8246	0.8235	0.7988
12	0.6764	0.7843	0.6957	0.9301	0.9288	0.7681	25	0.7803	0.7747	0.7745	0.8180	0.8125	0.9623
13	0.6856	0.7028	0.6898	0.8492	0.8504	0.7818	average relative percentage error : $(\sum_{i=1}^{5}(\lvert sim_i - sbp_i \rvert / \lvert sim_i \rvert)) / n$						0.002

3.2 The Engineering Example of the BP-Based Case Retrieval

A high-rise building, height: 72.1m; floors: 20; function: office building; fortification intensity: 9; site classification: Ⅲ. It is required to find the most similar building in the cases library.

A5=(72.1,20,2,2,2) is the goal case. The forecast values ascertained by improved BP-based case retrieval are in Table 1. It is obviously that this

building is the most similar to the No.9's whose first selecting structure is RC.+shearing force. This selection is the same with the practice engineering. Therefore, this method can provide intelligent case retrieval for high-rise structure form optimization.

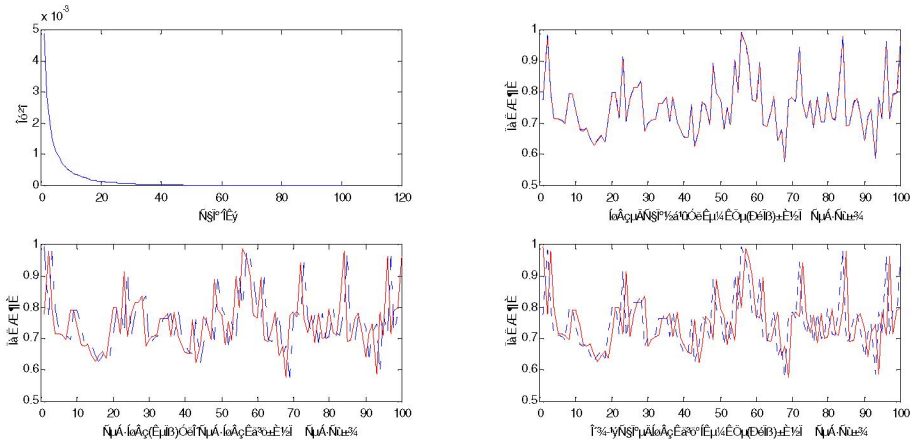


Fig.2 Training results of case-inference BP neutral network

4 Conclusion

First, we provided the process of CBR for high-rise structure form selection. Second, we analyzed the defect of the typical distance-based similarity measure method, i.e. it fails to measure the influence of attributes' weight, so when dimensions between attributes are not in uniform and their distances are widely different, contribution of part of the attributes to the similarity as a whole of the cases will be overlooked. Therefore, this paper established an improved BP-based intelligent CRM. It used mixed CRM to ascertain the similarity measure result of four target cases and twenty-five source cases. On the basis of that, it trained the BP network and provided the case retrieval examples based on this BP network. The result shows: this method can overcome shortcomings of traditional CRM such as poor self-learning, low efficiency and intelligent level, etc, providing a comparatively advanced means for high-rise structural form selection case retrieval.

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